Byung Hyun Lee

AI Researcher

Selective Knowledge Updates in Foundation Models

Expanding, Erasing, and Refining Knowledge with Minimal Forgetting

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KEY HIGHLIGHTS

RESEARCH SPECTRUM

Selective Knowledge Updates of Foundation Models

- **Expanding, removing and refining** knowledge while **minimizing forgetting**
- Both **post-training** (continual learning, unlearning, self-distillation & RL) and **training-free, closed-form updates** (GLoCE, DCAT)

Model & Domain Coverage

- **Diffusion models**, **multimodal LLMs**, **LLM reasoning**, **VLA models**, **pathology foundation models**
- **Gigapixel whole slide image** analysis, **industrial defect segmentation**, robot grasping and other **domain-specialized AI**

KEY HIGHLIGHTS ACHIEVEMENTS & EXPERIENCE

8 **Top-tier AI papers**
NeurIPS · CVPR · ICCV · ICLR · AAAI

7 **First / co-first author**
among top-tier papers

4 **ICLR 2027 submissions**
3 as co-first author

7 **Industry & gov. projects**
3 as project leader

Research Recognition

- NeurIPS 2026 **Spotlight** · AAAI 2024 **Oral** · CVPR 2026 **Compute Transparency Champion Award**
- MICCAI 2025 REG **2nd place (team leader)** · CVPR 2026 **Outstanding Reviewer** · Qualcomm Fellowship **Finalist**

Projects

- **Project leader** — **SD Biosensor** · **MSIT · NIPA** (gigapixel histopathology whole slide image) · **KETI** (world VLA model)
- **Samsung DS** (defect segmentation) · **Samsung Research** (robot grasping) · **Ministry of Health and Welfare** (gigapixel whole slide image AI) · **US Air Force · IITP** (human-machine teaming)

Industry Experience

- **NAVER Cloud**, National AI Foundation Model Project — contributed to **HyperCLOVA X 8B Omni / 32B Think**

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06 Domain-specialized AI & Projects

07 Additional Research — Image Restoration & Efficient Vision

PROFILE



Byung Hyun Lee

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https://hyun1a.github.io

EDUCATION

Ph.D., Electrical & Computer Engineering

Seoul National University | 2021.03 – 2027.02 (expected)

B.S., Electrical Engineering

UNIST | 2015.03 – 2018.08 | GPA 4.04 / 4.30

EXPERIENCE

Visiting Student Researcher, NAVER Cloud

National AI Foundation Model Project | 2025.10 – 2026.02

RESEARCH INTERESTS

Selective Knowledge Updates

Continual Learning
Concept Erasure · Unlearning
Training-free Weight Updates
Post-training · RL

Models & Domains

Diffusion Models
Multimodal LLMs · VLA
LLM Reasoning
Pathology Foundation Models
Industrial Inspection

PUBLICATIONS

8
Top-tier conference
7 as first / co-first

4
Under review
ICLR 2027 · 3 co-first

4
Workshop / short
CVPR-W · ECCV-W · ICASSP

2
Journal papers
IEEE Access · PLOS One

2
Technical reports
HyperCLOVA X 8B · 32B

1
Patent
Granted in US · KR

PUBLICATIONS FIRST / CO-FIRST AUTHOR

CONFERENCE (SELECTED PAPERS)

- Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs **NeurIPS 2026 (Spotlight)***
- Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models **CVPR 2026***
- Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis **ICCV 2025**
- Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation **CVPR 2025***
- Concept Pinpoint Eraser for Text-to-Image Diffusion Models via Residual Attention Gate **ICLR 2025***
- Doubly Perturbed Task-Free Continual Learning **AAAI 2024 (Oral)**
- Online Continual Learning on Hierarchical Label Expansion **ICCV 2023***

* Equal contribution

UNDER REVIEW · PREPRINT

- HYPERION: Hyperbolic Grounding for Histopathological Slide Representations **ICLR 2027 submitted***
- Self-Distilled Reinforcement Learning for Language-Robust VLA Models **ICLR 2027 submitted***
- S3-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation **ICLR 2027 submitted***
- Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis **Preprint 2026***

JOURNAL

- Towards Accelerating Model Parallelism in Distributed Deep Learning Systems **PLOS One 2023***

SHORT PAPER · DOMESTIC

- Empirically Accelerating Scaled Gradient Projection Using DNN for Inverse Problems **ICASSP 2021**
- Selective Concept Erasing for Safe Diffusion Models (Best Poster Award) **KSPC 2024**

PUBLICATIONS CO-AUTHOR · TECHNICAL REPORTS · PATENTS

CONFERENCE · UNDER REVIEW

- All-in-one Image Restoration for Unknown Degradations Using Adaptive Discriminative Filters for Specific Degradations **CVPR 2023**
- RePath: Zero-Shot Whole-Slide Tumor Segmentation via Intra-Slide Relational Affinity **ICLR 2027 (sub.)**
- Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation **arXiv 2025**

JOURNAL

- Continual Test-Time Adaptation for Robust Remote Photoplethysmography Estimation **IEEE Access 2025**

TECHNICAL REPORT · HYPERCLOVA X TEAM

- HyperCLOVA X 8B Omni / 32B Think **2026**

WORKSHOP · DOMESTIC

- Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models **CVPR Workshop 2026**
- Expert Classifier Ensemble-based Post-processing Correction for Unbiased Scene Graph Generation **IPIU 2024**
- Mobile AI & AIM 2022 Challenge Reports (quantized super-resolution; depth estimation) **ECCV Workshop 2022**
- Uncertainty-based Dual-Domain Low-Dose X-ray CT Reconstruction **KSPC 2022**

PATENT

- All-in-one image quality improvement model **US 2026 · KR 2025 (granted)**

PROJECTS & AWARDS

PROJECTS

KETI (Project Leader) — **2026 – present**
Uncertainty-aware World VLA Model

MSIT · NIPA (Project Leader) — **2026**
Hyperbolic Pathology FM for Gigapixel WSI

SD Biosensor (Project Leader) — **2025**
Industry–academia collaboration

US Air Force · IITP — **2025 – present**
AI for Human-Machine Teaming

Samsung DS — **2024 – 2025**
Industrial Defect Segmentation

Ministry of Health and Welfare — **2023 – 2024**
AI System for Gigapixel Histopathological WSI Analysis

Samsung Research — **2022 – 2023**
Robot Grasping

AWARDS & HONORS

CVPR Compute Transparency Champion Award — **2026**
Erasing Thousands of Concepts (ETC)

CVPR Outstanding Reviewer (Top 5%) — **2026**

Qualcomm Innovation Fellowship Korea — **2025**
Best Paper Finalist

MICCAI REG Challenge — 2nd Place (Team Leader) — **2025**
Report generation from gigapixel WSIs · 26 teams

Apple Scholars in AI/ML PhD Fellowship — **2025**
Institutional Nominee

Yulchon AI Star Scholarship — **2025**

KSPC Best Poster Presentation Award — **2024**

ACADEMIC SERVICE

Reviewer: ICLR · NeurIPS · CVPR · ECCV · AAAI · IEEE TIP · PR

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WHY KNOWLEDGE UPDATES

Foundation models now power products on three fronts — and each must keep updating what it knows after deployment.

Agentic / Physical AI

LLM agents · VLA-driven robots

Online knowledge addition

Adapt quickly to new tasks, tools, environments and instructions

Vertical AI

Medical imaging · industrial inspection

Domain knowledge addition

New organs, cancer types, processes and defect types keep appearing

Visual Generative AI

Text-to-image / video generation

Harmful knowledge deletion

Remove copyrighted, private or unsafe concepts on demand

INDUSTRY

Deployed models should adapt to new environments — without full retraining

- **Online knowledge addition** agents & robots
- **Domain knowledge addition** medical & industrial AI

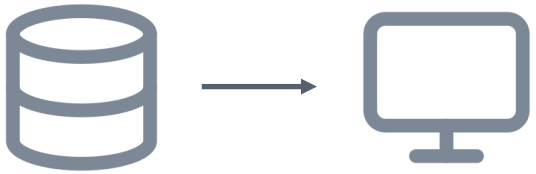
RESEARCH

- **Catastrophic forgetting** in foundation models
- **Stability–plasticity** trade-off
- **Precise, scalable unlearning**

THE CHALLENGE: Update knowledge without forgetting

1 Full retraining

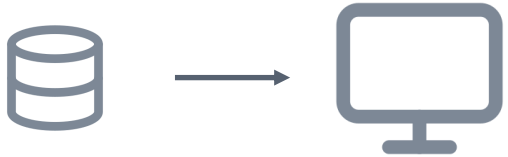
Retrain with all available data



Highly inefficient and costly

2 Naive knowledge update

Fine-tune with only the data to add or remove



Forgets existing knowledge (stability–plasticity dilemma)

3 Selective knowledge update

Update with only the target data while preserving existing knowledge



existing knowledge kept

Efficient and minimal forgetting

RESEARCH OVERVIEW

Selective Knowledge Updates in Foundation Models

Expanding, Erasing, and Refining Knowledge with Minimal Forgetting

Expanding

Continually add new knowledge

CONTINUAL LEARNING

HLE	ICCV 2023
DPCL	AAAI 2024 Oral
CoMEL	ICCV 2025

Erasing

Remove harmful or sensitive data

CONCEPT ERASURE · UNLEARNING

CPE	ICLR 2025
GLoCE	CVPR 2025
ETC	CVPR 2026

Refining

Merge and refine existing knowledge

MODEL MERGING · POST-TRAINING

DCAT	NeurIPS 2026 Spotlight
S³-OPSD	ICLR 2027 sub.
MOSAR (VLA)	ICLR 2027 sub.
HYPERION	ICLR 2027 sub.

APPLICATION

Visual Generative AI | Agentic · Physical AI | Medical AI (Pathology) | Industrial · Semiconductor AI

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Online Continual Learning on Hierarchical Label Expansion

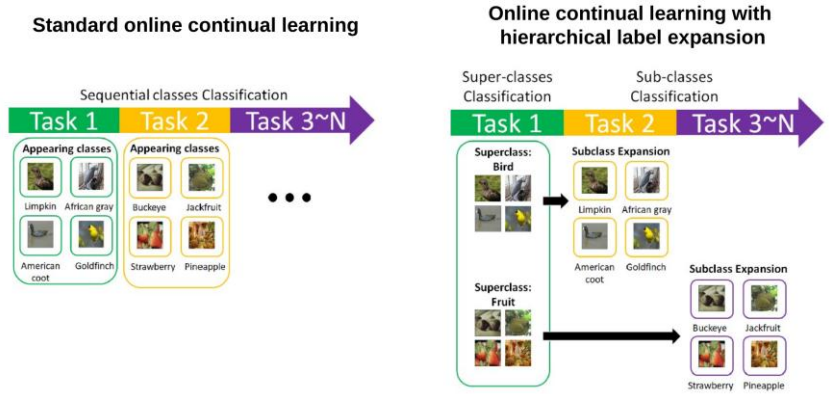
ICCV 2023 | Co-first author

EXPANDING

Keywords *online continual learning, hierarchical labels, memory management, any-time inference*

TASK OBJECTIVE

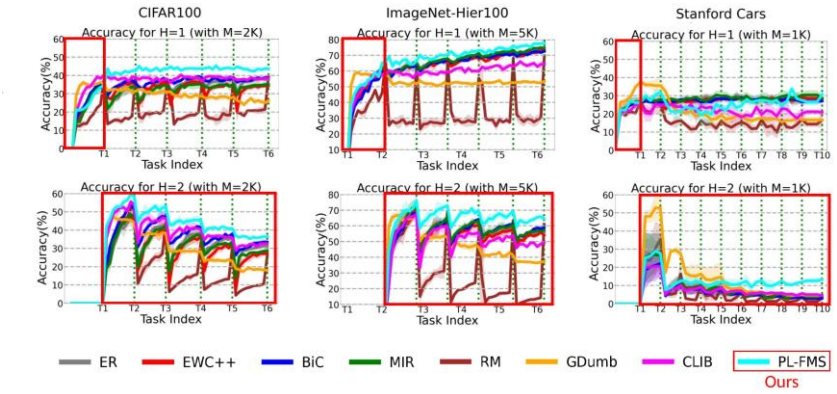
1. Define **hierarchical label expansion (HLE)** for **online continual learning**: classes are refined from coarse to fine as data streams in
2. Learn new **fine-grained classes** while retaining coarse-level knowledge



Standard online CL (left) vs. hierarchical label expansion (right)

RESEARCH SUMMARY

- New online CL setting reflecting real class hierarchies, with **single- and multi-depth** scenarios
- **Pseudo-label-based memory management** that estimates hierarchy branches and weighs samples per branch
- **Flexible Memory Sampling** that adapts the memory-to-new-data ratio over time
- **Consistently best** any-time accuracy on CIFAR100, ImageNet-Hier100 and Stanford Cars



Any-time inference (top: super-class, bottom: sub-class)

Doubly Perturbed Task-Free Continual Learning

AAAI 2024 · Oral | First author

EXPANDING

Keywords *task-free continual learning, future samples, input & weight perturbation, flat minima*

TASK OBJECTIVE

1. Go beyond preserving past samples: **account for future samples** in online continual learning
2. **Improve stability and plasticity together** with a theoretically grounded objective

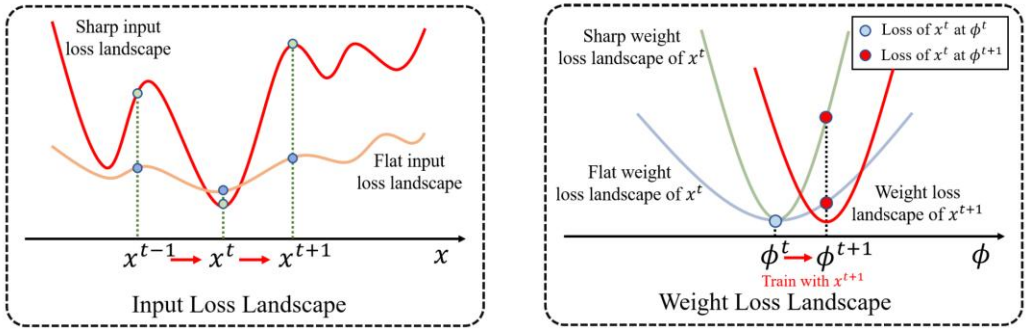
RESEARCH SUMMARY

- Proved the future-aware loss is **upper-bounded by input- and weight-perturbation terms**
- **Perturbed Function Interpolation** (feature noise + mixup) for the input term
- **Branched Stochastic Classifiers** (SWA + variational inference) for the weight term
- **Best any-time accuracy** on CIFAR100, CIFAR100-SC and ImageNet-100, with flatter loss landscapes

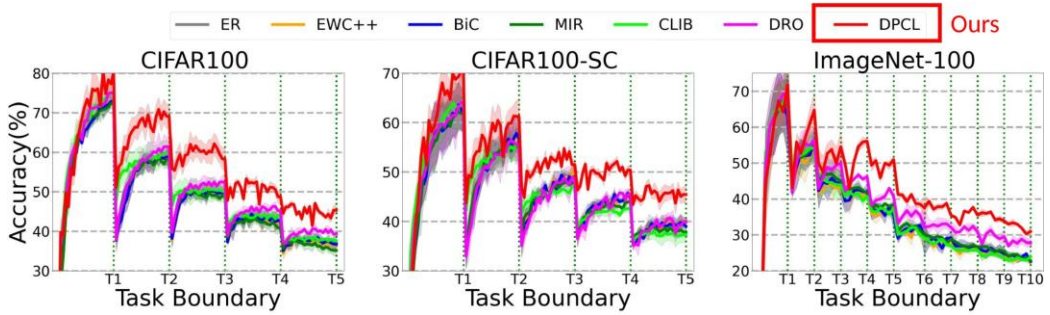
$$\mathcal{L}_t(\theta) + \lambda \max_{\|\Delta x\| \le \eta_1^t} \mathcal{L}_{t,\Delta}(\theta) + \rho \max_{\|\Delta \phi\| \le \eta_2^t} \max_{\|\Delta x\| \le \eta_1^t} \mathcal{L}_{t,\Delta}([\theta_e; \phi^t + \Delta\phi])$$

Input perturbation

Weight perturbation



Upper bound of the future-aware loss; input / weight loss landscapes



Any-time inference accuracy in class-incremental learning

Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis

ICCV 2025 | First author

EXPANDING

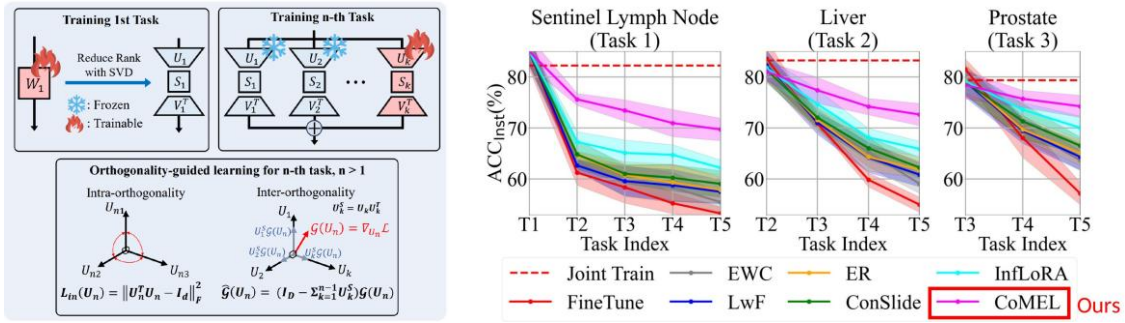
Keywords *whole slide image, multiple instance learning, memory-free CL, LoRA, localization*

TASK OBJECTIVE

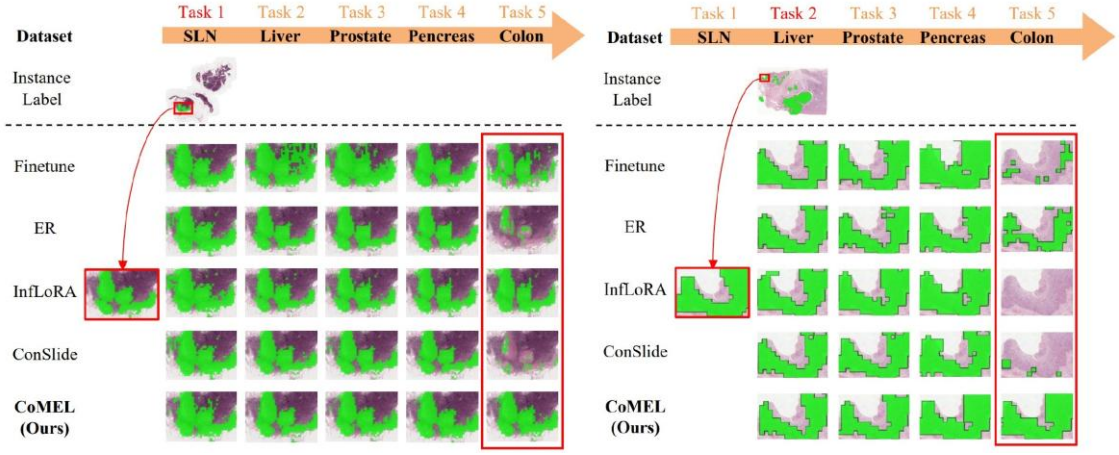
1. Continual learning for gigapixel **whole slide images** as new organs and cancer types arrive
2. **Memory-free**: keep both **diagnosis and lesion localization** without storing patient data

RESEARCH SUMMARY

- Two gaps in prior WSI continual learning: **replay-memory privacy risk** and **localization forgetting**
- **Orthogonal Weighted LoRA**: new bases orthogonal to previous tasks minimize interference
- **Bag Prototype-based Pseudo-Labeling** sharpens patch-level localization
- **Least localization forgetting** across five sequential organs (SLN → Liver → Prostate → Pancreas → Colon)



OWLoRA (left) and per-organ localization accuracy across tasks (right)



Tumor localization over the task sequence (CoMEL: ours)

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Concept Pinpoint Eraser for Text-to-Image Diffusion Models via Residual Attention Gate

ICLR 2025 | Co-first author

ERASING

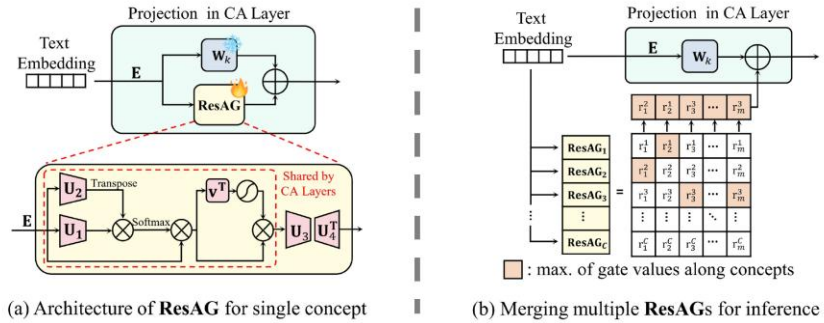
Keywords *concept erasure, text-to-image diffusion, residual attention gate, AI safety*

TASK OBJECTIVE

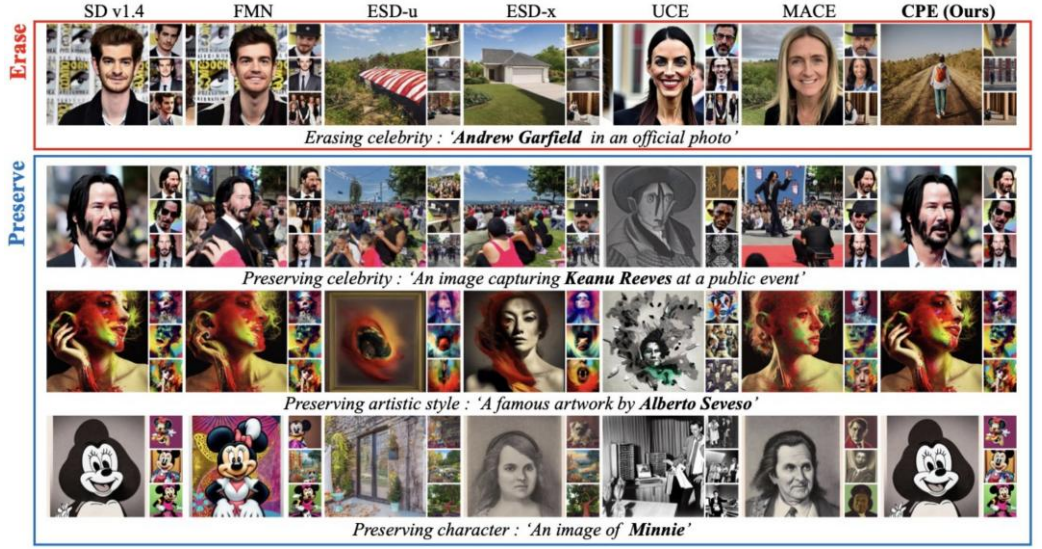
1. **Erase target concepts** (portrait rights, copyright) from text-to-image diffusion models
2. **Preserve all remaining concepts** — minimal forgetting during erasure

RESEARCH SUMMARY

- Attention-only fine-tuning cannot tighten the output-change bound
  → an **erasing–preservation dilemma**
- **Residual Attention Gate (ResAG)**: an add-on module that activates only for target concepts
- Per-concept ResAGs merge in parallel for **multi-concept erasure**
- **Near-original fidelity** on remaining concepts while erasing 50 celebrities / 100 styles



ResAG (left) and merging multiple ResAGs (right)



Erasing 50 celebrities: erased vs. preserved concepts

Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation

CVPR 2025 | Co-first author

ERASING

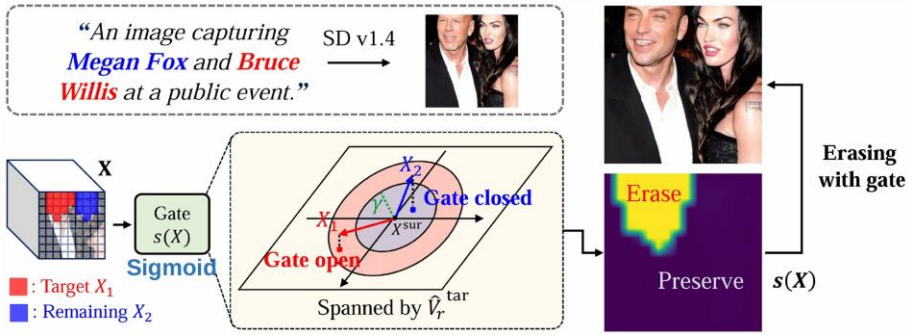
Keywords *localized concept erasure, training-free, low-rank adaptation, gating*

TASK OBJECTIVE

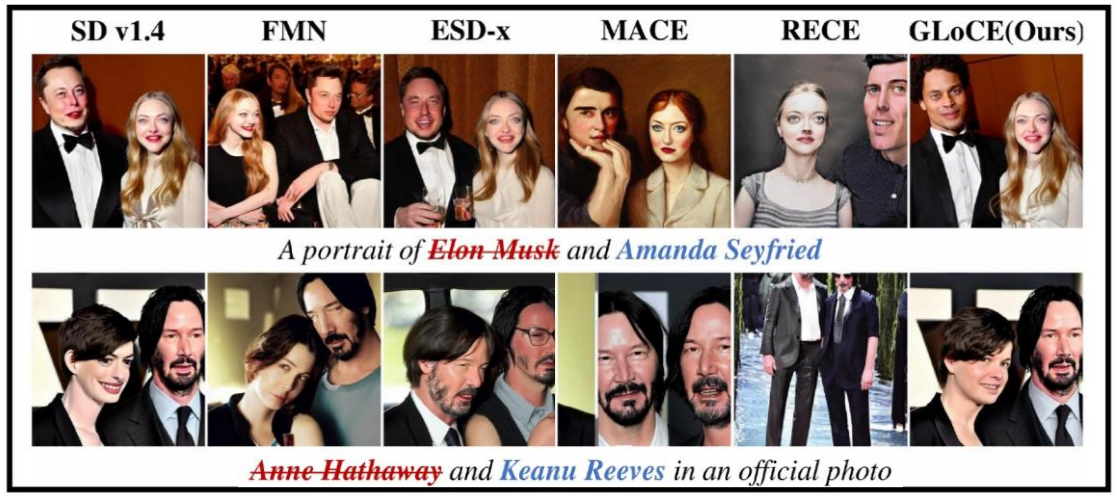
1. When several concepts share an image, **remove only the target concept's region**
2. **Keep the rest of the image intact** — without additional training

RESEARCH SUMMARY

- Concept image tokens concentrate in a **low-rank subspace**
- **Training-free low-rank update**: remove target components, project onto a mapping concept
- A **gate** that opens only on target regions → **GLoCE**, an inference-only update
- Localized erasure on **SD v1.4 / SD v3**, preserving other identities and content



Gate mechanism: erasure only on target-concept regions



Localized celebrity erasure

Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models

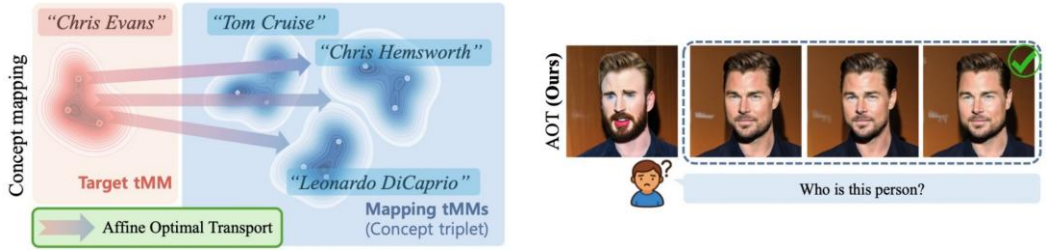
CVPR 2026 | Co-first author · Compute Transparency Champion Award

ERASING

Keywords *scalable concept erasure, optimal transport, mixture-of-experts, robustness*

TASK OBJECTIVE

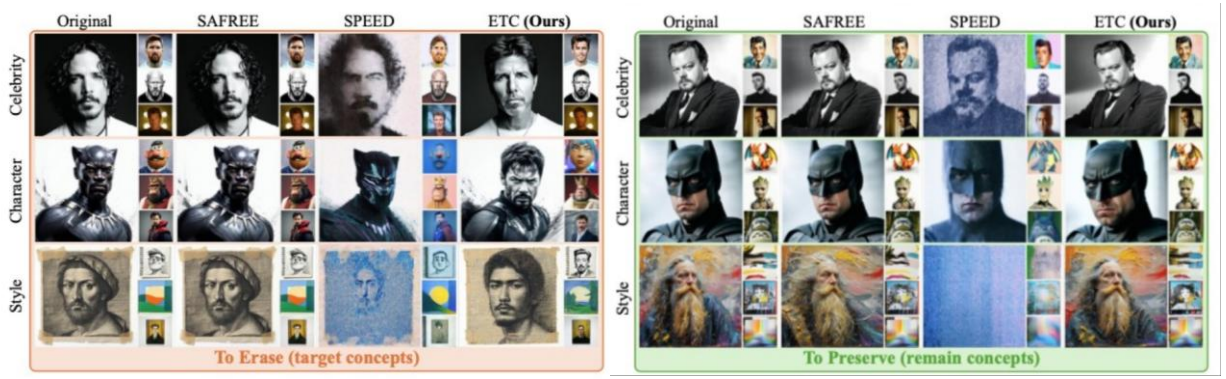
1. Scale concept erasure to **thousands of concepts** across celebrities, styles and characters
2. **Preserve quality and remaining concepts**, with **robustness** to white-box attacks



Concept mapping via affine optimal transport

RESEARCH SUMMARY

- Concept embeddings as **t-mixtures**, mapped target → mapping via **affine optimal transport**
- **Boundary anchoring** preserves neighbors without predefined anchor concepts
- **MoEraser**: one MoE module erases all targets; noise-injected training resists module removal
- Erased **2,072 concepts on SD v1.4** and **515 on SD v3.5** at once



MORE ON GENERATIVE MODELS

Unlearning · Semantic Binding · Safe Diffusion

Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models

CVPR Workshop 2026 · Co-author

- Removes outputs that **prompts cannot specify**, e.g., a particular person's face
- **Image-editing surrogates** + timestep-aware weighting + gradient surgery
- Validated on **SD3** and **DDPM-CelebA**

Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation

arXiv 2025 · Co-author

- Attributes in multi-object prompts often **bind to the wrong object**
- Analyzes the **geometry of text-token embeddings** behind attribute–object binding
- Uses it to **strengthen semantic binding** in text-to-image generation

Selective Concept Erasing for Safe Diffusion Models

KSPC 2024 · First author · Best Poster

- Erases **unsafe concepts** while keeping unrelated generations intact
- Received the **Best Poster Award**
- Early work that grew into **CPE, GLoCE and ETC**

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Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs

NeurIPS 2026 · Spotlight | Co-first author · DCAT

REFINING

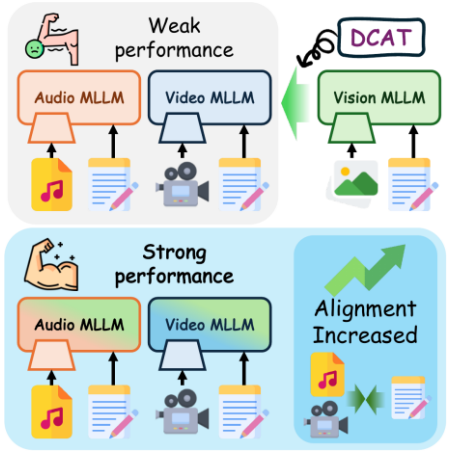
Keywords *multimodal LLM, model merging, cross-modal alignment, closed-form update*

TASK OBJECTIVE

1. Improving a weak MLLM modality (**audio, video**) usually needs large modality-specific data and compute
2. Transfer alignment from a **strong donor** (vision) to a **weak recipient** — **no fine-tuning**

RESEARCH SUMMARY

- Merging a well-aligned donor MLLM **improves the recipient on its own benchmarks**
- A **mutual-information lower bound** defines an **Alignment score** that tracks performance
- **DCAT**: a **closed-form, layer-wise weight update** from a few hundred calibration samples
- **+29.55% (audio)** and **+24.92% (video)** on average; up to **+100.92%**

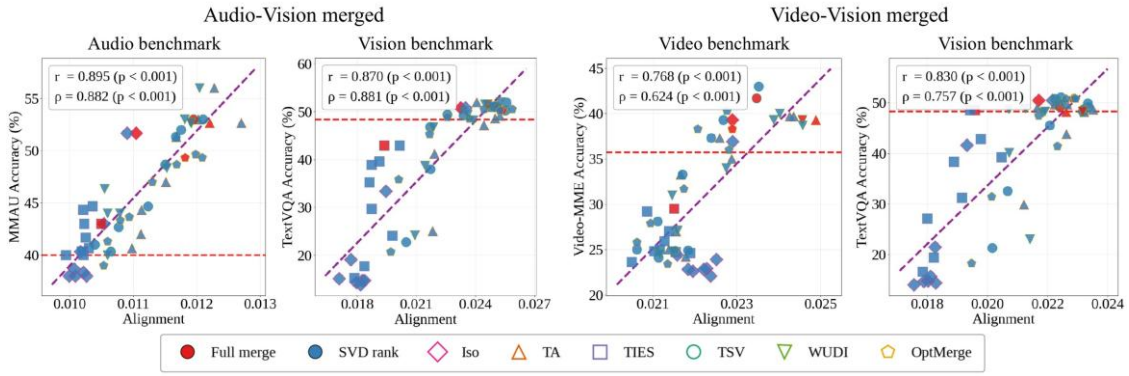


DCAT concept

+29.55% Avg. gain, audio benchmarks

+24.92% Avg. gain, video benchmarks

r = 0.895 Alignment–accuracy correlation



Alignment score vs. modality performance across merged checkpoints

S³-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation

ICLR 2027 (submitted) | Co-first author

REFINING

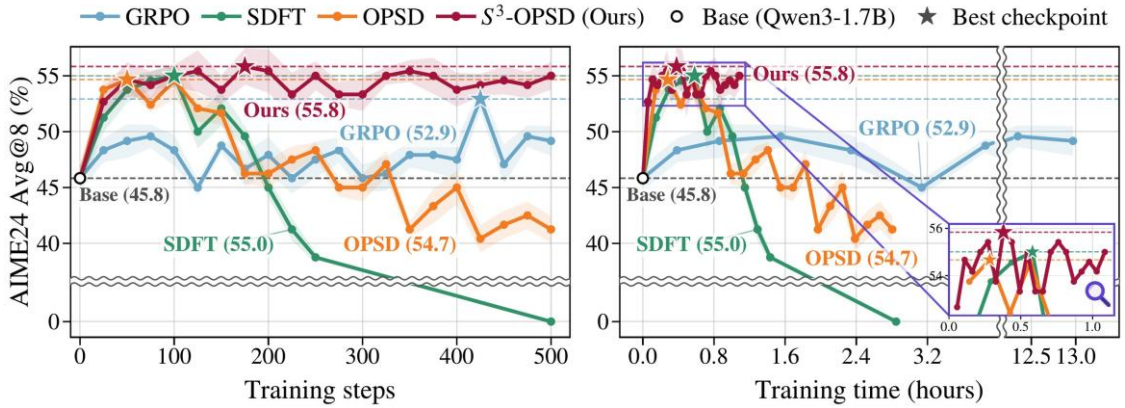
Keywords *LLM reasoning, on-policy self-distillation, sparse supervision, stability*

TASK OBJECTIVE

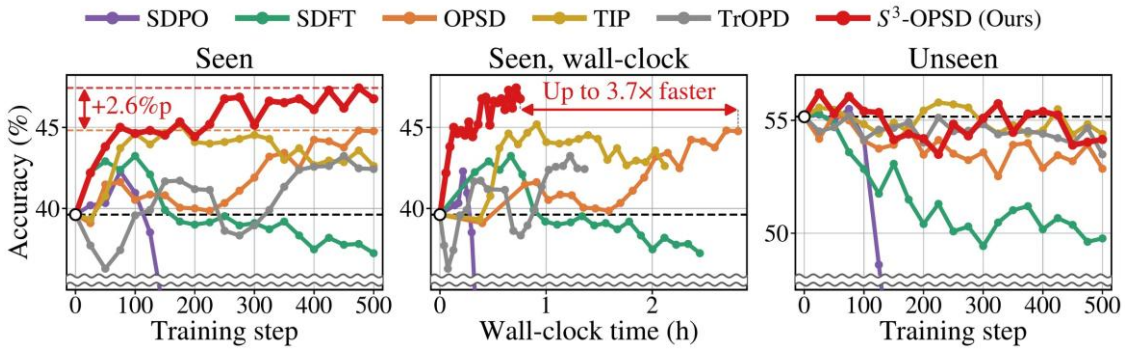
1. **On-policy self-distillation (OPSD)** improves LLM reasoning without an external teacher, but is **costly and unstable**
2. Make OPSD **sparse, stable and fast** across model families and tasks

RESEARCH SUMMARY

- **Sparse token selection** by advantage variance, via a cheap two-sided-bounded proxy
- **SGD** gives sparse updates and less drift than momentum optimizers → stability
- **Target smoothing** with the initial student as a fixed reference
- AIME24 **55.8 (+10.0 vs. base)**; up to **13× faster than GRPO**, **3× than OPSD**



AIME24 of Qwen3-1.7B vs. training steps (left) and wall-clock time (right)



Multi-task learning over seven domains

Self-Distilled Reinforcement Learning for Language-Robust VLA Models

ICLR 2027 (submitted) | Co-first author · MOSAR

REFINING

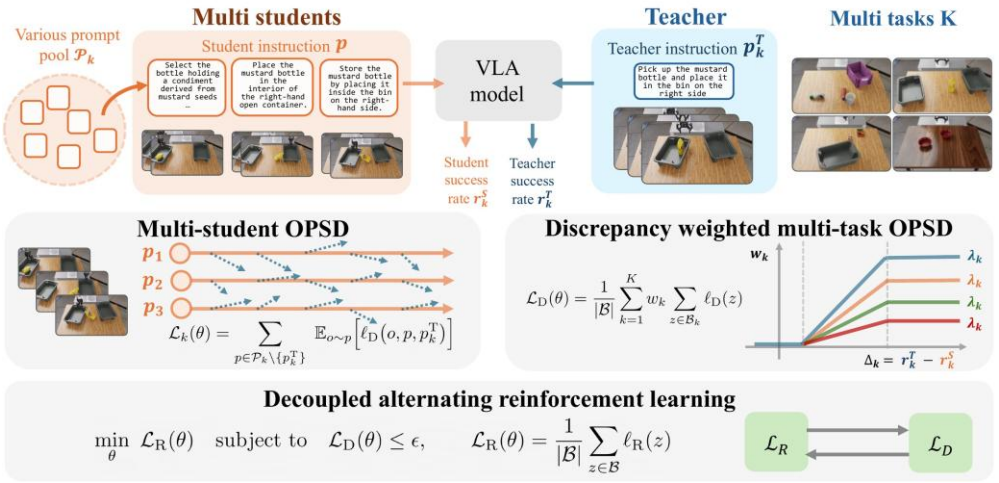
Keywords *vision-language-action, reinforcement learning, self-distillation, robotics*

TASK OBJECTIVE

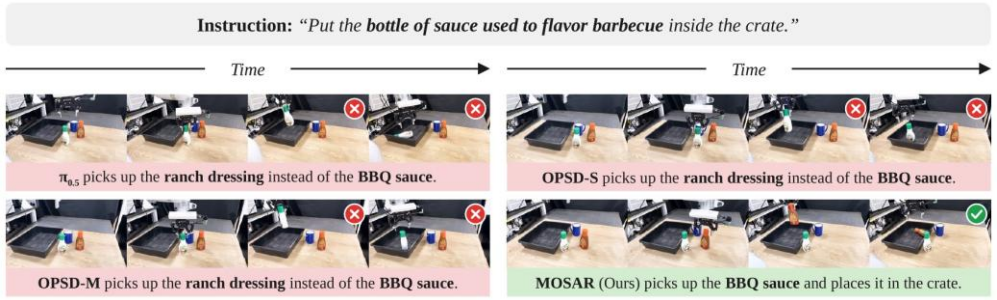
1. VLA policies are **sensitive to prompt phrasing**: paraphrases yield very different success rates
2. Close the gap, **exceed the best phrasing**, and keep unseen-task performance

RESEARCH SUMMARY

- **Multi-student OPSD**: the best prompt teaches paraphrased prompts at student-visited states
- **Alternating RL and distillation**: RL raises the ceiling, distillation spreads it
- **Discrepancy-weighted** distillation by per-task success gaps
- Unseen prompts **53.2% vs. 33.7% (PPO)**; REALM **61.0% vs. 56.8%**; **real Franka robot**



Overview of MOSAR



Real-robot evaluation: only MOSAR picks the correct object

INDUSTRY EXPERIENCE NAVER Cloud · HyperCLOVA X

Visiting Student Researcher

NAVER Cloud | Oct 2025 – Feb 2026

Korea's National AI Foundation Model Project

~2×
faster inference, kept stable

+20%
training tokens per unit time

ROLE & CONTRIBUTIONS

- Post-training of **HyperCLOVA X**: improved **training & inference stability**
- Kept the model stable under **~2× faster inference**; built and optimized code for **online deployment**
- **Dynamic batching** → **+20%** tokens trained per unit time

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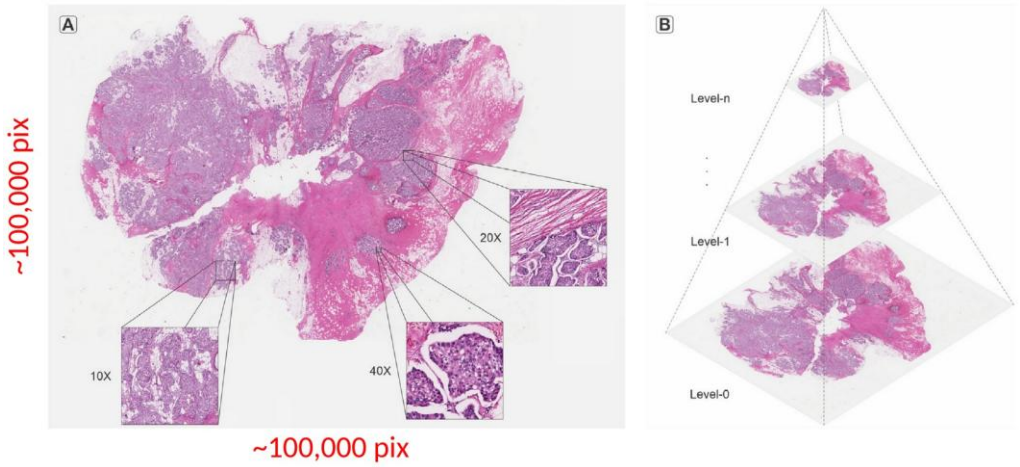
05 Refining — Model Merging and Post-training of Foundation Models

06 Domain-specialized AI & Projects

07 Additional Research — Image Restoration & Efficient Vision

GIGAPIXEL PATHOLOGY AI

Medical · Domain-specialized AI



Whole slide image: ~100,000 × 100,000 px

DOMAIN CHALLENGES

- Gigapixel inputs with only **slide-level labels**
- **Localizing** the evidence behind each diagnosis
- New organs and cancer types; patient-data **privacy**
- **Inference cost** of thousands of slides per day

ICCV 2025 First author

CoMEL — Continual MIL with Localization

Memory-free continual learning that keeps diagnosis and localization

ICLR 2027 sub. Co-first author

HYPERION — Hyperbolic Patch–Slide VLM

Best Dice on all five zero-shot segmentation benchmarks

Preprint 2026 Co-first author

Speculative Encoding

Up to 3× faster, 6× less memory, >99% performance retained

MICCAI 2025 2nd place · team leader

REG 2025 — Tree-of-Experts Report Generation

Structured reports from gigapixel WSIs (26 teams)

ICLR 2027 sub. Co-author

RePath — Zero-shot Tumor Segmentation

Intra-slide relational affinity for zero-shot segmentation

HYPERION: Hyperbolic Grounding for Histopathological Slide Representations

ICLR 2027 (submitted) | Co-first author

DOMAIN-SPECIALIZED AI

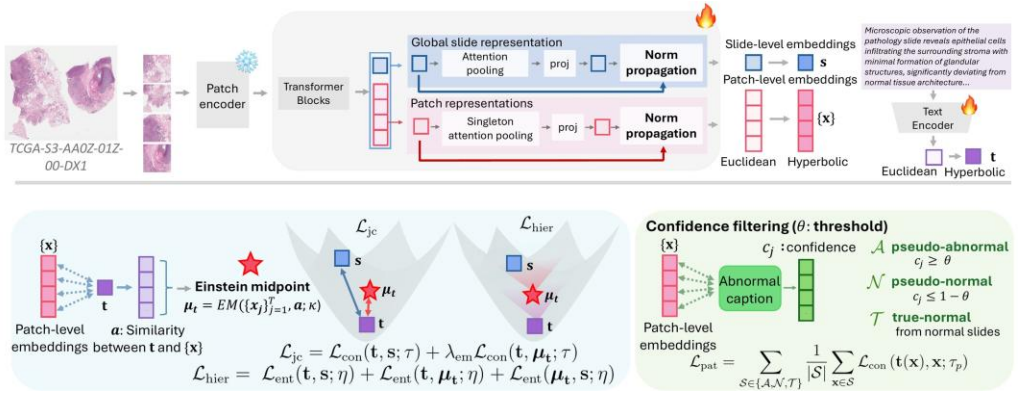
Keywords *computational pathology, hyperbolic VLM, zero-shot segmentation, report generation*

TASK OBJECTIVE

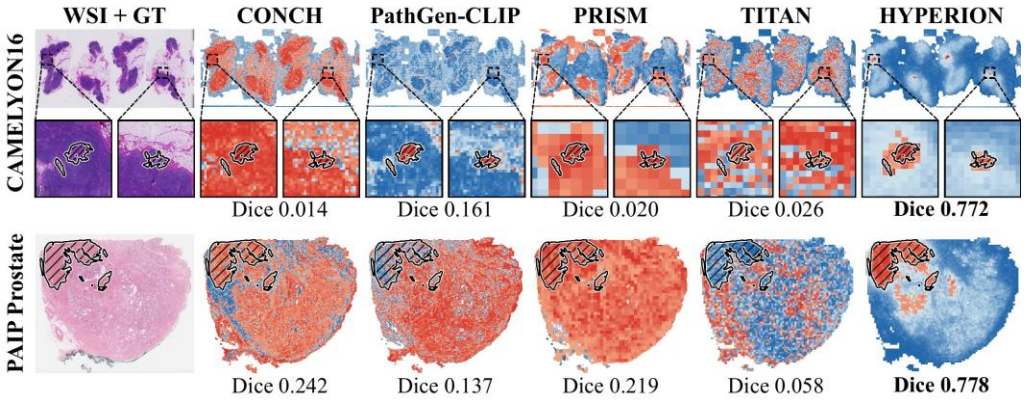
1. Euclidean slide foundation models **discard feature-norm information** that could encode hierarchy and fine detail
2. **Ground pathological concepts in patch-level evidence** for patch–slide representation

RESEARCH SUMMARY

- **Hyperbolic VLM** with contrastive alignment and **entailment cones**
- **Text-conditioned Einstein midpoints** + confidence-filtered **patch pseudo-labels**
- **Norm propagation** exposes intermediate feature norms in patch radii
- **Best Dice on all 5 benchmarks** (0.143 of baseline → 0.463); knowledge kept on **LLM tasks and prediction tasks**



Overview of HYPERION



Zero-shot tumor segmentation vs. pathology foundation models

Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis

Preprint 2026 | Co-first author

DOMAIN-SPECIALIZED AI

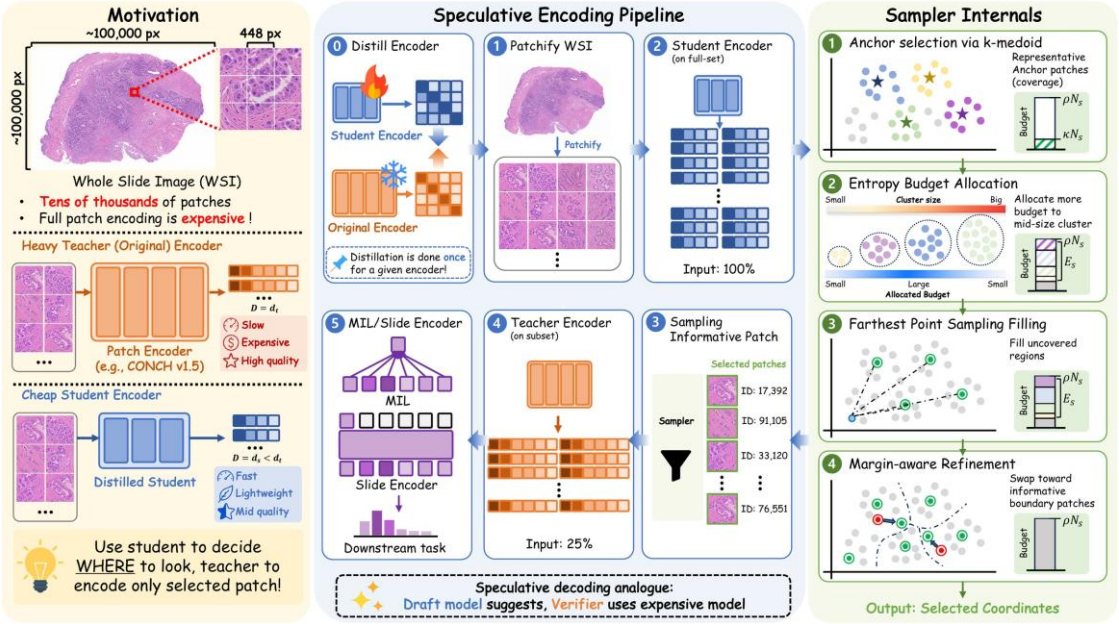
Keywords *efficient inference, whole slide image, distillation, patch sampling*

TASK OBJECTIVE

1. WSI inference is costly: **patch encoding dominates compute**, **slide aggregation drives peak memory**
2. Hospitals produce **thousands of slides per day** → fast, memory-efficient **inference** is essential

RESEARCH SUMMARY

- MIL models and slide encoders **retain performance with a sparse subset** of patches
- A **distilled light encoder drafts** features; the **full encoder runs only on selected patches**
- Sampler: k-medoid anchors, entropy budgets, farthest-point filling, margin-aware refinement
- **Up to 3× faster**, **6× lower peak memory**, **>99%** of performance (9 MIL models, 3 slide encoders)



Draft with a cheap distilled encoder, verify with the full encoder

3×
faster MIL inference
4.63 s → 1.16 s

6×
lower peak memory
TITAN: 10.4 → 1.6 GB

>99%
performance kept
9 MIL · 3 slide encoders

Tree-of-Experts for Report Generation in Histopathology

MICCAI 2025 · REG Challenge | 2nd of 26 teams · Team leader

DOMAIN-SPECIALIZED AI

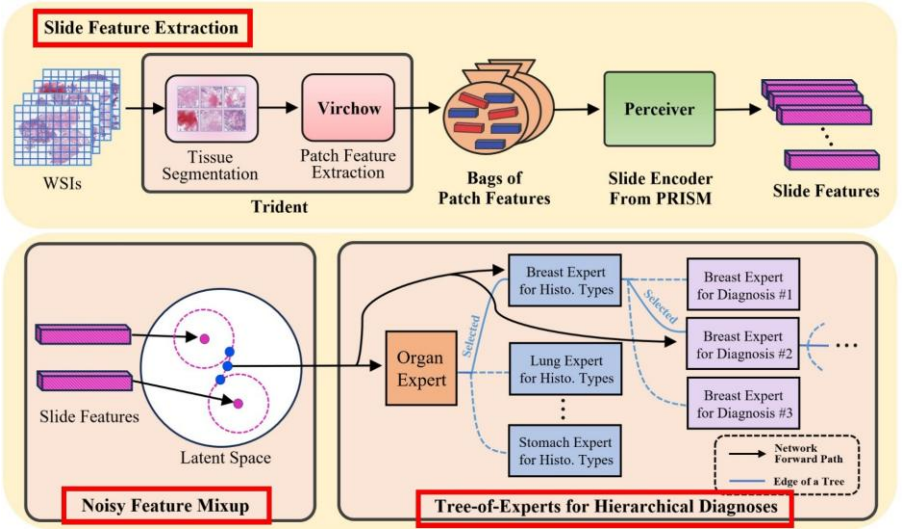
Keywords *report generation, gigapixel WSI, slide foundation model, feature augmentation*

TASK OBJECTIVE

1. Generate **structured pathology reports** (organ → procedure → histologic type → diagnoses)
2. Stay **robust across eight organs** with fewer than 10K training slides

RESEARCH SUMMARY

- Reframed generation as **hierarchical structured prediction**
- **Tree-of-Experts**: **64 lightweight MLP experts** predict report components
- **Frozen slide features** (GrandQC → Virchow → PRISM), extracted once
- **Noisy Feature Mixup** for data-scarce experts; ranking score **0.811** across 8 organs



Slide features, Noisy Feature Mixup and Tree-of-Experts

	Breast	Lung	Colon	Rectum	Stomach	Cervix	Prostate	Bladder	Avg.
Ranking	0.775	0.842	0.840	0.710	0.871	0.774	0.801	0.798	**0.811**
ROUGE	0.793	0.884	0.845	0.762	0.855	0.838	0.806	0.875	**0.832**

Validation scores per organ

Training-free Few-shot Industrial Defect Segmentation

Samsung DS | Industry project · Dec 2024 – Mar 2025 DOMAIN-SPECIALIZED AI

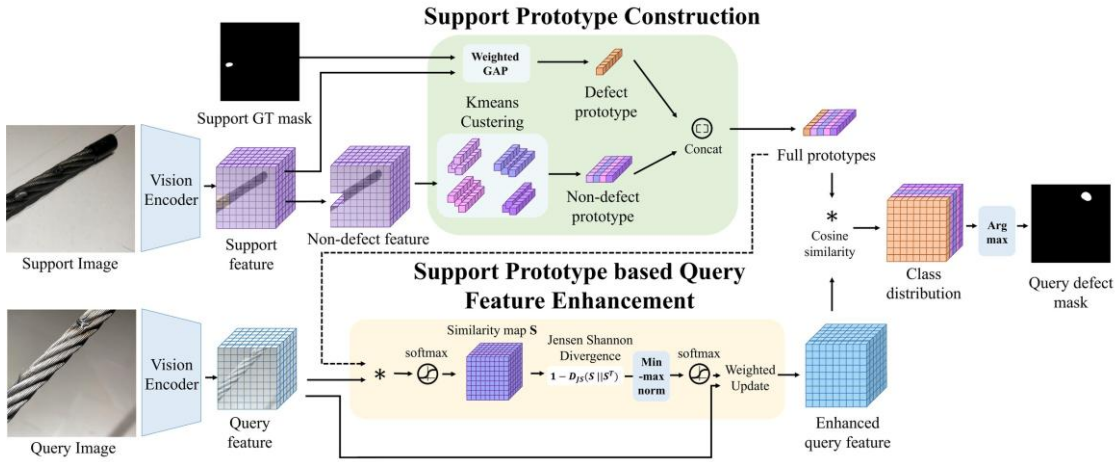
Keywords *industrial inspection, few-shot segmentation, training-free, DINOv2*

TASK OBJECTIVE

1. Segment **specific defect types** from **one annotated example** — anomaly detection cannot tell types apart
2. Avoid **training on costly masks** and handle **unseen (OOD) defects**

RESEARCH SUMMARY

- **Training-free** pipeline on a frozen **DINOv2** encoder
- **Defect + non-defect prototypes** (masked pooling, k-means) suppress false positives
- **Query enhancement** by Jensen–Shannon similarity to support prototypes
- **Best training-free mIoU**: VISION V1 **22.0** vs. 11.1 (SegGPT), on par with training-based SOTA



Support prototypes and prototype-based query enhancement

	Method	VISION V1	Defect Spectrum
Training	SOFS (SOTA)	22.5	39.1
Training-free	SegGPT	11.1	31.5
Training-free	**Ours**	**22.0**	**36.5**

1-shot defect segmentation, mIoU (%)

PROJECTS & LEADERSHIP

Industry · Government · Challenge

AS PROJECT LEADER / TEAM LEADER

SD Biosensor
Project Leader · 2025

Industry–Academia Collaboration

Industrial

MICCAI 2025 REG
Team Leader · 2nd / 25

Report Generation from Gigapixel WSIs

Medical

MSIT · NIPA
Project Leader · 2026

GPU Support: Pathology FM for Gigapixel WSI

Medical

KETI
Project Leader · 2026 –

Uncertainty-aware World Model ↔ VLA Interaction

Physical · Agentic

AS PARTICIPATING RESEARCHER

Samsung DS
2024 – 2025

Training-free Few-shot Industrial Defect Segmentation

Industrial · Semiconductor

Samsung Research
2022 – 2023

Relation & Affordance Aware Robot Grasping

Physical

Korea MOHW
2023 – 2024

Gigapixel Histopathology WSI AI System

Medical

US Air Force · IITP
2025 – present

AI for Human-Machine Teaming

Agentic

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IMAGE RESTORATION & EFFICIENT VISION

Low-level vision · On-device AI

CVPR 2023 · Co-author

All-in-one Image Restoration for Unknown Degradations Using Adaptive Discriminative Filters

- A single model restores images with **unknown degradation types** via degradation-specific adaptive filters
- Granted **US (2026) and KR (2025) patents**

ECCV Workshop 2022 · Challenge participant

Mobile AI & AIM 2022 Challenges — Quantized Super-Resolution on Mobile NPUs · Depth Estimation on Mobile

- Participated in **on-device** challenges: **quantized super-resolution on mobile NPUs** and **depth estimation on mobile devices**
- Co-author of the official **challenge reports**

ICASSP 2021 · First author

Empirically Accelerating Scaled Gradient Projection Using Deep Neural Network for Inverse Problems

- Accelerates iterative **inverse-problem solvers** for image processing with a learned network

KSPC 2022 · Co-author

Uncertainty-based Dual-Domain Low-Dose X-ray CT Reconstruction

- **Uncertainty-based**, **dual-domain** reconstruction for low-dose X-ray CT

CONCLUSION

01 Selective Knowledge Update Expertise

Updating what foundation models know — **adding, removing and refining** knowledge while preserving the rest — through both **post-training** (continual learning, unlearning, self-distillation with RL) and **training-free, closed-form updates** (GLoCE, DCAT)

02 From Problem Definition to Large-scale Validation

Realistic problem to **large-scale validation** (2,000+ concepts, gigapixel WSIs, MLLMs, VLAs on a real robot)

03 Leadership & Industry Collaboration

Project leader for SD Biosensor, MSIT · NIPA and KETI projects; **team leader** for MICCAI 2025 REG (2nd place); contributed to **NAVER Cloud HyperCLOVA X**; projects with **Samsung DS and Samsung Research**

Thank you

Byung Hyun Lee · AI Researcher

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